



FIFA World Cup Ticket Alert — Full Project Plan

A website where fans get **notified the moment tickets become available** for the matches they care about (new releases **and** official resale), so they can buy them **themselves**. We are an **alerting / tracking** service.

🚫 **We do NOT buy tickets for users, auto-checkout, or create multiple accounts.** Those are illegal under anti-bot ticketing laws (US BOTS Act, UK, EU). This product is the legal, fair alternative — and a real business.

1. The idea in one line

"Be the first to know when World Cup tickets drop — for your matches, on your phone, instantly."

2. Who it's for

- Fans who keep missing tickets because they sell out in minutes.
- People watching for **official resale** tickets to appear.
- Fans who want alerts only for **their team / their city / their budget**.

3. Core features

#	Feature	Description
1	Sign up / log in	Email + password (or Google login)
2	Browse matches	All World Cup matches, with venue, date, status
3	Watchlist	User picks which matches + categories + max price to watch
4	Availability monitor	Background checker watches official + resale sources
5	Instant alerts	Web Push + Email (and optionally Telegram/WhatsApp)
6	"Buy now" link	Alert links straight to the official purchase page
7	Alert history	See past alerts so you know what you missed
8	Subscription (optional)	Free tier (1–2 matches) + paid tier (all matches, faster checks)
9	Admin panel	Add/edit matches and data sources, see system health

4. Tech stack (recommended)

Layer	Choice	Why
Frontend + backend	Next.js (already in this repo)	One framework, React UI + API routes

Layer	Choice	Why
Ticket data	Ticketmaster Discovery API (first-party) + SeatGeek API (resale)	Real APIs with free tiers — see section 5
Database	PostgreSQL via Supabase or Neon (free tier)	Real relational DB for users, watchlists, alerts
Auth	Supabase Auth or NextAuth.js	Handles signup/login/Google securely
Cache / fast state	Upstash Redis	Last-seen ticket status, rate-limit counters
Notifications – push	Web Push (already built)	Instant browser/phone alerts
Notifications – email	Resend or SendGrid	Backup channel, free tiers
Notifications – chat (optional)	Telegram Bot API (free) / WhatsApp	Power users love instant chat alerts
Scheduler	Vercel Cron (Pro) or cron-job.org (free)	Triggers the availability checks
Payments (optional)	Stripe	Subscriptions for the paid tier
Hosting	Vercel (free HTTPS)	Web Push requires HTTPS
Error monitoring	Sentry (free tier)	Know when checks fail

5. Where the data comes from (most important section)

There is **no official FIFA tickets API**, but there are **real ticketing APIs** we can use — so we don't rely on fragile scraping. The legal line stays simple:

 **Reading availability (via APIs) and alerting humans = legal.**  **Auto-buying, auto-login, or fake accounts = illegal bot.**

Data sources, in order of preference:

#	Source	Covers	Free tier	Notes
1	Ticketmaster Discovery API	Official first-party tickets — FIFA's US venue partner, most of the 78 US matches	~5,000 req/day	<code>GET app.ticketmaster.com/discovery/v2/events.json</code> · key at developer.ticketmaster.com
2	SeatGeek Platform API	Resale / secondary market (StubHub, Vivid Seats...) + fair-price score	Yes	Great for "resale appeared" + "is the price fair" alerts
3	FIFA official waitlist / app	First-party drops	—	If FIFA exposes a "notify me" feature, mirror it
4	Public ticket pages (fallback)	Anything the APIs miss	—	Only if needed, slowly + respecting robots.txt

Rules we follow:

- Use the **APIs** first; keep within their rate limits.
- **Never** log in, add to cart, or buy. We only read availability.
- Always send the user to the **official purchase page** to complete the buy.
- Respect Ticketmaster/SeatGeek terms — they forbid bot buying; we only read + alert.
- Show a clear disclaimer: we are an independent alert service, not FIFA.

6. Database schema (starter)

```
users
  id, email, password_hash (or oauth), tier (free|paid), created_at

matches
  id, name, stage, venue, city, kickoff_at, ticket_url, status

watchlist
  id, user_id → users, match_id → matches,
  category (e.g. Cat1/Cat2/Cat3), max_price, active

alerts          (history of what we sent)
  id, user_id, match_id, type (NEW_DROP|RESALE), sent_at, channel

push_subscriptions
  id, user_id, endpoint, keys_json
```

Fast-changing state lives in Redis:

```
status:{matchId}      = "AVAILABLE" | "SOLD_OUT"      // last seen
checkLock:{matchId}   = lock so two crons don't double-check
```

7. How a single alert flows (end to end)

```
1. Cron triggers /api/check-tickets
2. For each match with active watchers:
   call Ticketmaster (+ SeatGeek) API → AVAILABLE or SOLD_OUT
3. If it changed SOLD_OUT → AVAILABLE:
   find all users watching that match (+ matching category/price)
   send them Web Push + Email (+ Telegram if enabled)
   record rows in `alerts`
4. Save new status in Redis (so we don't re-alert next run)
```

De-dupe rule: only alert on a **change** to AVAILABLE, and at most once per match per X minutes per user.

8. Build phases (do them in order)

Phase	What you build	Result
0. MVP	Follow ALERT.md	Push alert when any watched match has tickets
1. Accounts	Add auth + per-user watchlists (Postgres)	Each user watches their own matches
2. Email	Add Resend/SendGrid as a 2nd channel	Alerts also hit the inbox
3. Categories/price	Filter alerts by category + max price	Fewer, more relevant alerts
4. Resale source	Add official resale monitoring	Catch resale drops too
5. Paid tier	Stripe subscription, faster checks for payers	Revenue
6. Telegram/WhatsApp	Add chat alerts	Power users
7. Admin + monitoring	Admin panel + Sentry	Run it reliably

9. Costs (rough, to start)

Service	Free tier	When you pay
Vercel	Yes	Pro (\$20/mo) for built-in minute-cron
Supabase/Neon (Postgres)	Yes	When you grow past free rows
Upstash Redis	Yes	High volume
Resend/SendGrid email	Yes (limited/day)	More emails
cron-job.org	Yes	—
Stripe	Pay per transaction	Only when you charge users

You can launch the MVP for \$0.

10. Legal / compliance checklist (do not skip)

- ☐ We only **read public availability** and **alert** — never buy.
- ☐ No multiple/fake accounts, no checkout automation.
- ☐ Clear disclaimer: "Independent alert service. Not affiliated with FIFA."
- ☐ Respect each source's robots.txt / terms; throttle our checks.
- ☐ Privacy policy for storing user emails/subscriptions (GDPR-friendly).
- ☐ If charging money, clear terms of service + refund policy.

11. Definition of done (v1 public launch)

- ☐ Users can sign up, log in, and pick matches to watch
- ☐ Background checker runs reliably on a schedule
- ☐ Users get Web Push + Email when a watched match has tickets

- ☐ Alert links to the official purchase page
- ☐ No duplicate spam; alert history visible
- ☐ Legal disclaimer + privacy policy live